Introduction to E-Portfolio Complex Systems

Lecture 3: Properties of Complex Dynamical Systems

Properties of Complex Dynamical Systems in Beijing Air Quality Dynamics (2010–2014)

Group ePortfolio Entry · Complex Systems · Tilburg University

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Introduction

This portfolio entry builds on Module 2 (Temporal Dynamics and Change) that examined how PM2.5 pollution in Beijing changed over time between 2010 and 2014. According to Kelso (1995), the principles of self-organisation underpin all pattern formation in nature, and this entry investigates whether that holds for Beijing's pollution system. The dataset, containing 43,824 hourly observations, is based on Liang et al. (2015), who recorded PM2.5 which are fine airborne particles that directly reflect air quality, alongside variables such as temperature, pressure, and wind speed. Missing PM2.5 values (n = 2,067) were filled using linear interpolation prior to analysis. In Module 2, we established that the PM2.5 time series has memory, exhibits regime shifts, and is dynamically structured. This entry investigates what generates those patterns, specifically through the lens of self-organisation, emergence, and the conditions that give rise to complex dynamics. Analyses were conducted in Python using pandas, matplotlib, seaborn. The statistical analyses were done in R, such as the Bartels rank test, KPSS, and TV-AR.
Dataset : Beijing PM2.5, [UCI Machine Learning Repository]

```
# Load dataset
df = pd.read_csv('PRSA_data_2010.1.1-2014.12.31.csv')
# Fill missing PM2.5 values using linear interpolation (n=2,067)
 df['pm2.5'] = df['pm2.5'].interpolate(method='linear')
# Encode wind direction as numeric
le = LabelEncoder() df['cbwd'] = le.fit_transform(df['cbwd'])
```

This entry addresses the following questions:

1. In what ways does the system manage the problem of complexity of substance and problem of pattern complexity?
2. Does the system being studied exhibit a bifurcation?
3. Does the data exhibit complex dynamics, and what are the scientific and practical implications of that?

In what ways does the system manage the problem of complexity of substance and problem of pattern complexity?

Substance complexity explains how a pattern emerges from many interacting components working together (Kelso, 1995). In Beijing's pollution system, variables such as temperature, atmospheric pressure, dew point, and wind speed all interact to explain the PM2.5 levels. Rather than any single variable driving pollution, the pattern emerges from the simultaneous interaction of all components, which is the defining feature of substance complexity.

```
# Select PM2.5 and meteorological variables
dataframe = df[['pm2.5', 'DEWP', 'TEMP', 'PRES', 'Iws', 'Is', 'Ir', 'cbwd']] corr_matrix =
dataframe.corr()
# Plot correlation heatmap — warmer = positive, cooler = negative
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f") plt.title('Correlation
Between PM2.5 and Weather Variables (2010-2014)')
plt.show()
```

Figure 1:

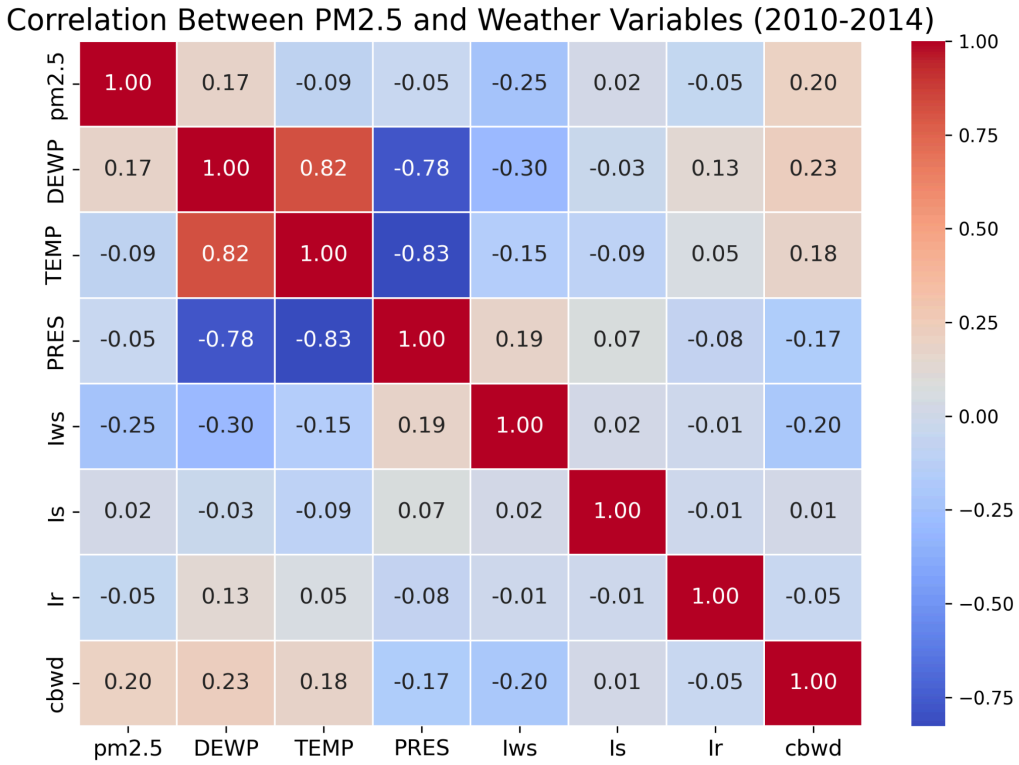


Figure 1 shows the correlation between PM2.5 and the relevant variables across 2010-2014.

No single variable correlates strongly with PM2.5 across 2010 to 2014. The strongest relationships are wind speed (Iws = -0.25) and wind direction (cbwd = 0.20). This suggests that pollution levels are driven by multiple interacting factors, rather than one dominant cause, consistent with Kelso's (1995) notion of substance complexity.

Pattern complexity refers to how the same system can produce different patterns depending on conditions (Kelso, 1995).

```
df.boxplot(column='pm2.5', by='month', figsize=(12,6))
plt.title('PM2.5 Distribution by Month (2010-2014)')
plt.xlabel('Month')
plt.ylabel('PM2.5 (µg/m³)')
plt.xticks(range(1,13),
       ['Jan','Feb','Mar','Apr','May','Jun',
       'Jul','Aug','Sep','Oct','Nov','Dec'])
plt.show()
```

Figure 2: PM2.5 Distribution by Month (2010-2014)

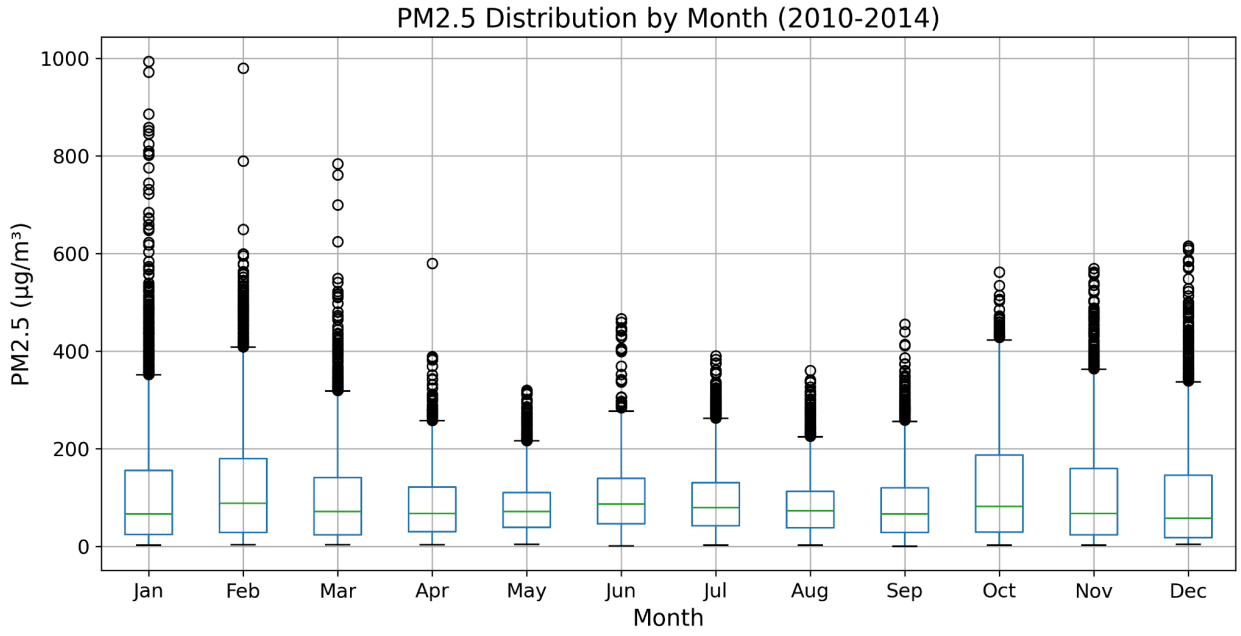


Figure 2 shows the PM2.5 distribution across months. Winter months like January and February show much higher pollution peaks compared to summer months, while summer months are comparatively calm and stable. The same atmospheric system, driven by the same underlying variables, produces fundamentally different pollution patterns depending on seasonal conditions, which is precisely what pattern complexity describes (Kelso, 1995).

Does the system being studied exhibit a bifurcation?

A bifurcation occurs when a small change in a parameter causes the system to suddenly shift between different states. It visualizes instability at a critical point. When the parameter passes this critical point, the system changes its behavior and oscillates between multiple states instead of staying at one. For real environmental data like PM2.5, constructing a classical bifurcation diagram is not straightforward since there is no single clear control parameter that drives the system. Instead, we look for bifurcation-like behaviour through critical slowing down. The idea is that a system's variance increases as it approaches a critical threshold (Olthof et al., 2020).
Figure 3 illustrates this behavior directly from the PM2.5 data, showing where the system shifts between stable and unstable states over time. It does not pinpoint exactly when shifts occur.

```
# Rolling standard deviation to detect bifurcation-like behaviour # A spike in variance suggests the system is approaching a critical threshold # Window = 720 hours = 30 days
df['datetime'] = pd.to_datetime(df[['year', 'month', 'day', 'hour']])
rolling_var = df['pm2.5'].rolling(window=720).std()

plt.figure(figsize=(12,6))
plt.plot(df['datetime'], rolling_var, color='steelblue')
plt.title('PM2.5 Rolling Standard Deviation (2010-2014)', fontsize=16)
plt.xlabel('Date', fontsize=14)
plt.ylabel('Rolling Std (PM2.5)', fontsize=14)
plt.show()
```

Figure 3: Rolling Standard Deviation of PM2.5

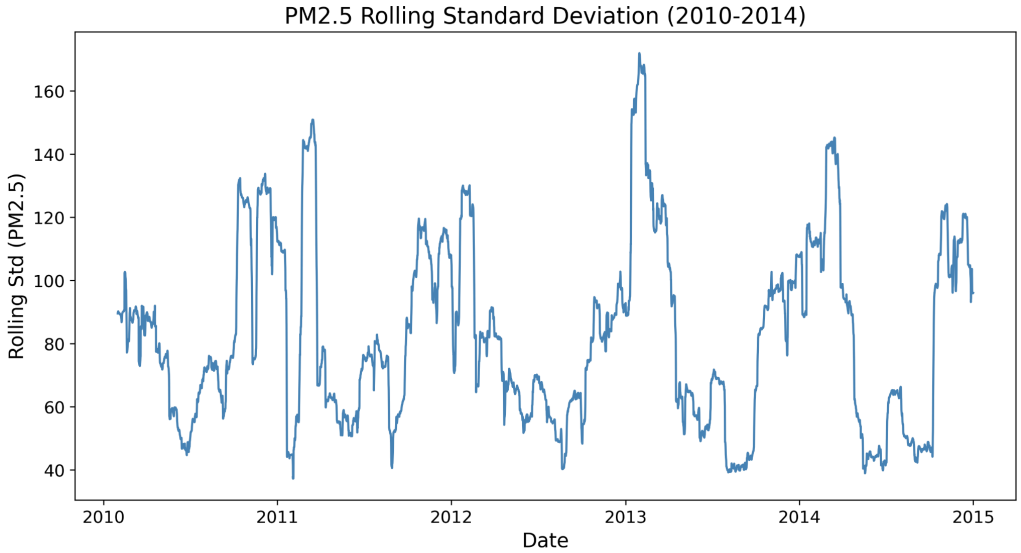


Figure 3 shows the rolling standard deviation of PM2.5 over time. The system switches between calm periods, where values drop to around 40-50, and unstable periods where it suddenly jumps to 150-170. These sudden shifts show the system crossing a critical point, which is consistent with bifurcation-like behaviour

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Does the data exhibit complex dynamics? And, what are the scientific and practical implications of that?

To determine whether data exhibit complex dynamics, three characteristics are required (Olthof et al., 2020).

1. Complex systems develop over time and possess memory. The current state depends on previous states
2. The system must show phase transitions between attractor states within the time series, which appear as regime shifts and are non-stationary.
3. Complex systems show path dependence and sensitivity to initial conditions. The future behavior depends on their past states.

As established in Module 2, the ACF of PM2.5 shows strong autocorrelation that slowly fades over approximately 100 hours. This confirms that today's pollution levels are still influenced by past states, meaning the system has memory. To further test this, a Bartels Rank Test in R, with the *'randtests'* package, was applied to PM2.5. The test returned a p-value of < 2.2e-16, which strongly confirms that the series is non-random and that past states influence future ones (Olthof et al., 2020).

```
randtests::bartels.rank.test(df$pm2.5, alternative ="two.sided")
```

Outcome:

```
Bartels Ratio Test

data:  df$pm2.5
statistic = -197.13, n = 41757, p-value < 2.2e-16
alternative hypothesis: nonrandomness
```

The KPSS test returns a p-value of 0.1, it suggests the series is stationary globally. However, KPSS tests for stationarity across the entire series; it does not detect local regime shifts within it.

```
library(tseries) kpss.test(na.omit(df$pm2.5))
```

Results KPSS:

- KPSS Statistic: 0.2576
- p-value: 0.1000

The rolling standard deviation plot clearly shows the system shifting between calm and unstable periods over time. This matches what Module 2 found, the system is statistically stationary overall, but KPSS does not detect local shifts. The system can be globally stable while still showing real regime shifts at specific points in time.

To further test whether the memory structure changes over time, a Time-Varying Autoregressive (TV-AR) model was applied to PM2.5. The model returned an EDF of 9.394. It is above 2, meaning the autocorrelation structure is not stable, it shifts across different periods. This suggests that how pollution levels relate to their past changes depending on the season or pollution episode, which is consistent with genuine complex dynamics. One limitation is that the results depend on the parameters chosen for the model, different settings could lead to slightly different outcomes.

```
# TV-AR model — tests whether memory structure changes over time
# EDF > 2 = non-stationary autocorrelation = genuine complexity
pm25_clean <- na.omit(df$pm2.5)
n <- length(pm25_clean)
tt <- 1:(n-1) tvar <- gam(pm25_clean[2:n]
      ~ s(tt, by=pm25_clean[1:(n-1)], k=10, bs="tp"))
stvar <- summary(tvar) round(stvar$edf, 3)
```

→ 9.394

```
# Plot the time-varying autoregressive parameter over time # Shows how the memory
structure of PM2.5 shifts across the observation period
plot.gam(tvar, select=1, rug=FALSE, seWithMean=TRUE,
       ylab="Autoregressive parameter",
       xlab="Time (hours)",
       main="Time-Varying AR Parameter of PM2.5 (2010-2014)")
```

Figure 4: Time-Varying AR Parameter of PM2.5 (2010-2014)

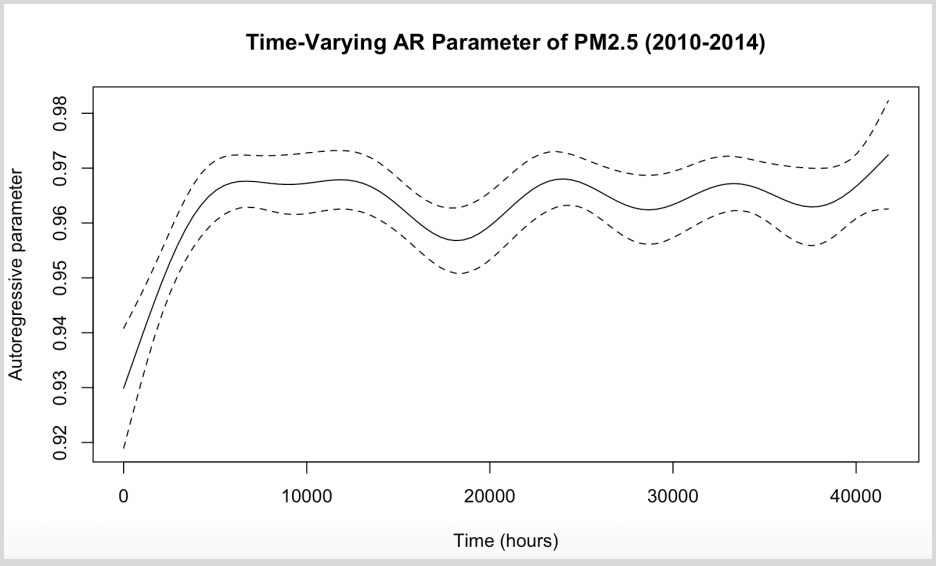


Figure 4 shows how the autoregressive parameter of PM2.5 changes over time. The parameter fluctuates between 0.93 and 0.97, meaning the memory structure is not stable and shifts depending on the period. This is consistent with an EDF of 9.394 and confirms genuine complex dynamics

```
# Filter January data across all years
# Same month, different years — tests for sensitive dependence on initial conditions
jan_data = df[df['month'] == 1]
jan_mean = jan_data.groupby('year')['pm2.5'].mean()

plt.figure(figsize=(10,6))
jan_mean.plot(kind='bar', color='steelblue', edgecolor='black')
plt.title('Average PM2.5 in January per Year (2010-2014)', fontsize=16)
plt.xlabel('Year', fontsize=14)
plt.ylabel('Mean PM2.5 (µg/m³)', fontsize=14)
plt.xticks(rotation=0, fontsize=12)
plt.yticks(fontsize=12)
plt.show()
```

Figure 5: Average PM2.5 in January per Year (2010-2014)

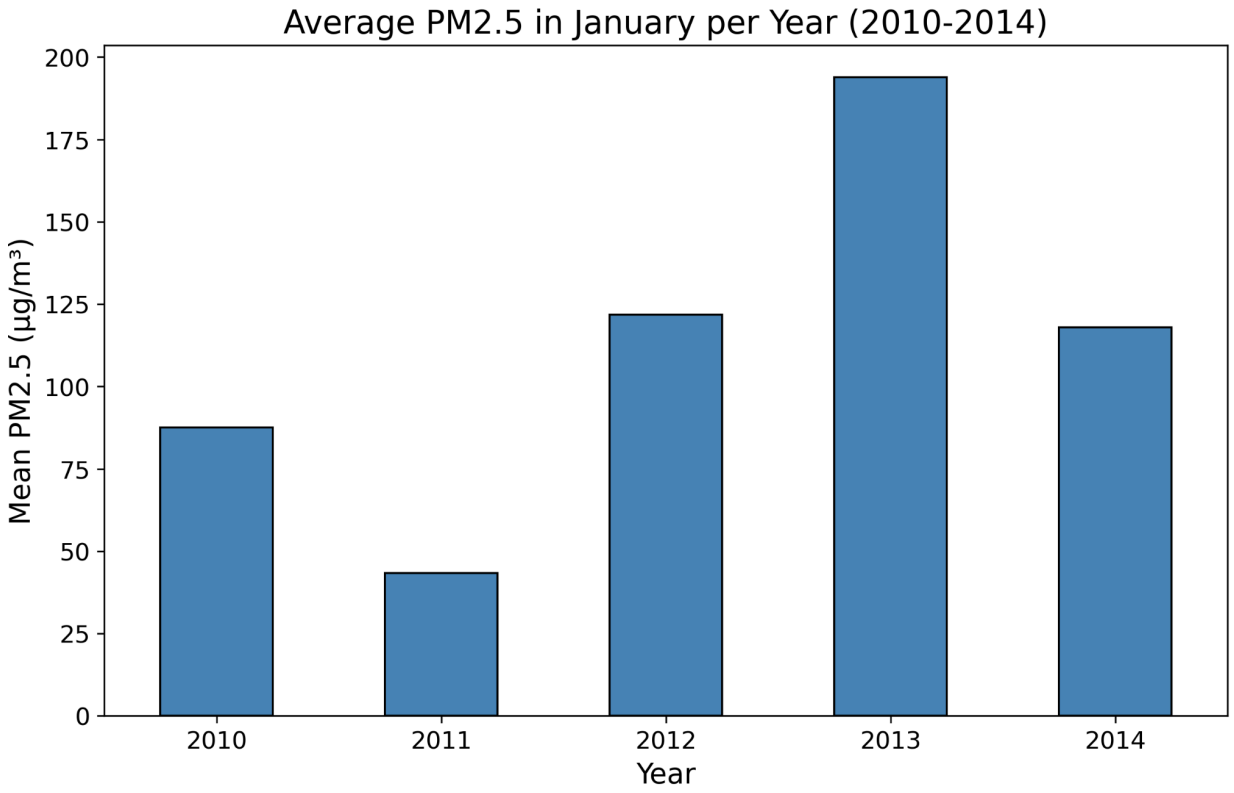


Figure 5 shows the average PM2.5 in January across different years. The same month produces very different pollution levels each year, ranging from around 45 in 2011 to nearly 195 in 2013. This indicates that the system is sensitive to its conditions, where small differences can lead to very different outcomes (Olthof et al., 2020).

The scientific and practical implications of complex dynamics are that it is only possible to predict a few time points ahead but not far into the future. Simple deterministic systems are predictable, random systems are not, complex systems are in the middle. For the dataset specifically, simple linear models won't capture the full dynamics, therefore there is a need for complexity-aware approaches. In a practical sense, predicting pollution levels far into the future is difficult. Extreme events like the 2013 peak are hard to predict in advance. Observing sudden increases in variance could work as an early warning signal for upcoming pollution episodes.

Should the Olthof characteristics be used like descriptive statistics before choosing a modeling approach?

The three complexity characteristics from Olthof et al. (2020) should be treated like descriptive statistics for complex systems. Before choosing any modeling approach, you should first check whether the data has memory, shows regime shifts, and is sensitive to initial conditions. For the PM2.5 dataset, all three were confirmed by statistical measures. This means simple linear models are not sufficient, the data needs complexity-aware approaches that can handle long memory, regime shifts, and non-linearity. Skipping this step means you might end up applying the wrong analytical approach to the data.

To summarize

This entry examined Beijing's PM2.5 through self-organisation and emergence. The correlation matrix showed that no single variable drives pollution alone, which is substance complexity. The seasonal box plot showed the same system behaving differently across seasons, which is pattern complexity. The rolling standard deviation showed the system shifting between stable and unstable states, consistent with bifurcation-like behavior.

The data also confirmed all three characteristics of complex dynamics (Olthof et al., 2020). The system has memory, confirmed by the Bartels Rank Test ($p < 2.2e-16$) and the TV-AR model (EDF = 9.394). It shows local regime shifts despite global stationarity, and is sensitive to its conditions. These findings build on Module 2, where PM2.5 was found to be statistically stationary but dynamically structured. This entry explains why the patterns emerge from many interacting weather variables working together.

Although this portfolio does not include a Module 4, it would be possible to explore this further by examining the attractor structure of the system and how it moves through its possible states over time.

References:

Kelso, J. A. (1995). How Nature Handles Complexity. In *Dynamic Patterns: The Self-Organization of Brain And Behavior*. The MIT Press.

Liang, Xuan, et al. "Assessing Beijing's PM2. 5 pollution: severity, weather impact, APEC and winter heating." *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 471.2182 (2015).

Olthof, M., Hasselman, F., & Lichtwarck-Aschoff, A. (2020). Complexity in Psychological Self-ratings: Implications for Research and Practice. *BMC Medicine*, *18*(1). https://doi.org/10.1186/s12916-020-01727-2

Links

China's pollution
https://colab.research.google.com/drive/11hTzmle7v8eYifMbgi1aKQ2dBuDumJW_#scrollTo=fhRHzoY6guas